

Remote Classroom Platform

1. Introduction

A Remote Classroom Platform is a web-based application that enables teachers and students to interact in real time from different locations. It provides features such as live video classes, real-time chat, and content sharing, making online learning interactive and engaging.

This platform aims to replicate the experience of a physical classroom in a virtual environment. Teachers can deliver lectures, share study materials, and conduct discussions, while students can actively participate, ask questions, and access recorded sessions for revision.

2. Tools and Technologies Used

Frontend Tools:

- HTML: Used to structure the web pages.
- CSS: Used for designing and styling the user interface.
- JavaScript: Used to add interactivity and dynamic behavior.

Backend Tools:

- Node.js: Used to handle server-side logic.
- Express.js: Used to manage server requests and routing.

Database:

- MongoDB / Firebase: Used to store user data, class details, and messages.

Video & Communication:

- WebRTC / Video APIs: Used for live video and audio communication.

Other Tools:

- Git & GitHub: Used for version control.
- VS Code: Used as the code editor.

3. Project Approach / Methodology

The development of the Remote Classroom Platform follows a systematic and modular approach:

1. Requirement Analysis:

- Identify user requirements for teachers and students.
- Define essential features such as video classes, chat, and content sharing.

2. System Design:

- Design the overall architecture of the system.
- Plan frontend layout and backend workflow.

3. Implementation:

- Develop frontend using HTML, CSS, and JavaScript.
- Implement backend services for authentication and data handling.
- Integrate video conferencing and chat features.

4. Testing:

- Test each module individually.
- Fix bugs and improve performance.

5. Deployment:

- Deploy the application on a web server.
- Ensure accessibility and security.

This structured approach helps in developing a reliable, efficient, and user-friendly remote classroom system.